L16

L16: The order of an integer continued ...

- We start this lecture with a sequence of results that establishes the number of primitive roots modulo m (under the assumption that at least one primitive root modulo m exists).

Proposition 5.4: Let $a, m \in \mathbb{Z}$ with $m > 0$ and $(a,m) = 1$. If i is a positive integer, then

$$\operatorname{ord}_m(a^i) = \frac{\operatorname{ord}_m(a)}{(\operatorname{ord}_m(a), i)}.$$

Proof: Put $d = (\operatorname{ord}_m a, i)$. Then there exist $b, c \in \mathbb{Z}$ such that $\operatorname{ord}_m a = db$, $i = dc$, and $(b, c) = 1$. Now,

$$(a^i)^b = (a^{dc})^{(\operatorname{ord}_m a)/d} = (a^c)^{\operatorname{ord}_m a} = (a^{\operatorname{ord}_m a})^c$$

$$\equiv 1 \pmod{m}.$$

By Proposition 5.1, we have that $\operatorname{ord}_m(a^i) \mid b$.

Also, $a^{i(\operatorname{ord}_m(a^i))} = (a^i)^{\operatorname{ord}_m(a^i)} \equiv 1 \pmod{m}$.

By Proposition 5.1, we have that ② $\operatorname{ord}_m a \mid i(\operatorname{ord}_m(a^i))$. Thus $db \mid dc(\operatorname{ord}_m(a^i))$, and we have that $b \mid c\operatorname{ord}_m(a^i)$. Since $(b,c)=1$, we obtain $b \mid \operatorname{ord}_m(a^i)$. Thus

$$\operatorname{ord}_m(a^i) = b = \frac{\operatorname{ord}_m a}{d} = \frac{\operatorname{ord}_m a}{(\operatorname{ord}_m a, i)},$$

as desired. ∎

Corollary 5.5: Let $a, m \in \mathbb{Z}$ with $m > 0$ and $(a,m) = 1$. If i is a positive integer, then $\operatorname{ord}_m(a^i) = \operatorname{ord}_m(a)$ if and only if $(\operatorname{ord}_m(a), i) = 1$.

Proof: Apply Proposition 5.4.

Corollary 5.6: If a primitive root modulo m exists, then there are exactly $\phi(\phi(m))$ incongruent primitive roots modulo m.

Proof: Let r be a primitive root modulo m.

Then $\mathrm{ord}_m r = \phi(m)$. By Proposition 5.3, each residue class coprime to m occurs exactly once in the set $\{r, r^2, r^3, \ldots, r^{\phi(m)}\}$. If i is a positive integer such that $1 \le i \le \phi(m)$, then $\mathrm{ord}_m(r^i) = \mathrm{ord}_m(r) = \phi(m)$ if and only if $(\phi(m), i) = 1$ by Corollary 5.5. The number of such integers i with $1 \le i \le \phi(m)$ is $\phi(\phi(m))$, as desired. $\square$

<u>Example</u>: • We showed in a previous lecture that 3 is a primitive root modulo 7. Thus there are exactly $\phi(\phi(7)) = \phi(6) = 2$ incongruent primitive roots modulo 7 by Corollary 5.6. By Corollary 5.5, $\mathrm{ord}_7(3^i) = \mathrm{ord}_7(3)$ if and only if $(\mathrm{ord}_7 3, i) = 1$. Since $\mathrm{ord}_7(3) = 6$, we must have $i = 1, 5$. Since $3^5 \equiv 5 \pmod{7}$, 5 is the other incongruent primitive root modulo 7.

- In a previous lecture we proved that 2 was a primitive root modulo 13. Thus there are exactly $\phi(\phi(13)) = \phi(12) = 4$ incongruent primitive roots modulo 13 by Corollary 5.6.

Exercise: Find the other primitive roots.

Remark: Consider the following question:
"Since $\phi(\phi(8)) = \phi(4) = 2$, do there exist two primitive roots modulo 8 by Corollary 5.6?"
The answer is no! We proved last lecture that there are no primitive roots modulo 8. Corollary 5.6 does not apply here!

Primitive Roots for Prime Numbers

- Our next goal is to prove that primitive roots exist for prime numbers. We will need Lagrange's Theorem for this. Lagrange's Theorem concerns the number of solutions a polynomial can have modulo p.

<u>Theorem 5.7</u> (Lagrange): Let p be a prime number and let ⑤

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

be a polynomial of degree $n \geq 1$ with integral coefficients such that $p \nmid a_n$. Then the congruence

$$f(x) \equiv 0 \pmod{p}$$

has at most n incongruent solutions modulo p.

<u>Proof</u>: We use induction on n. If $n = 1$, we have $f(x) = a_1 x + a_0$ with $p \nmid a_1$. Since $(p, a_1) = 1$, the linear congruence $a_1 x \equiv -a_0 \pmod{p}$ has exactly one incongruent solution modulo p by Theorem 2.6. Thus $f(x) \equiv 0 \pmod{p}$ has exactly one incongruent solution modulo p, and the Theorem holds for $n = 1$.

Assume that $k \geq 1$ and that the Theorem holds for polynomials of degree $n = k$. We must show that the Theorem holds for polynomials of degree

$n = k+1$. Accordingly, let

$$f(x) = a_{k+1}x^{k+1} + a_k x^k + \cdots + a_1 x + a_0$$

be a polynomial of degree $k+1$ with integral coefficients such that $p \nmid a_{k+1}$. If $f(x) \equiv 0 \pmod{p}$ has no solutions, then the Theorem holds and the proof is complete. Assume the congruence has at least one solution, say a. View $f(x) \in \mathbb{F}_p[x]$ ($\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a finite field so $\mathbb{F}_p[x]$ is a Euclidean domain / has a division algorithm). Thus

$$f(x) \equiv (x-a)q(x) + r \pmod{p},$$

where $q(x) \in \mathbb{F}_p[x]$ has degree k, leading coefficient not divisible by p, and $r \in \mathbb{F}_p$. Now,

$$\begin{aligned} f(a) &\equiv (a-a)q(a) + r \pmod{p} \\ &\equiv r \pmod{p}. \end{aligned}$$

Since $f(a) \equiv 0 \pmod{p}$, we have that $r \equiv 0 \pmod{p}$.

Thus $f(x) \equiv (x-a)q(x) \pmod{p}$.

Let b be any solution to $f(x) \equiv 0 \pmod{p}$.

Then $f(b) \equiv (b-a)q(b) \pmod{p}$.

If $b \not\equiv a \pmod{p}$, then $b-a \not\equiv 0 \pmod{p}$ and so $q(b) \equiv 0 \pmod{p}$. By the induction hypothesis, $q(x)$ has at most k incongruent solutions modulo p. Thus $f(x)$ has at most $k+1$ incongruent solutions modulo p, as required.

$\blacksquare$

<u>Proposition 5.8</u>: Let p be a prime and let $d \in \mathbb{Z}$ with $d > 0$ and $d \mid p-1$. Then the congruence

$$x^d - 1 \equiv 0 \pmod{p}$$

has exactly d incongruent solutions modulo p.

<u>Proof</u>: Since $d \mid p-1$, there exists $e \in \mathbb{Z}$ such that $p-1 = de$. Then

$$x^{p-1} - 1 = x^{de} - 1$$

$$= (x^d - 1)(x^{d(e-1)} + x^{d(e-2)} + \cdots + x^d + 1). \quad (8)$$

The congruence $x^{p-1} - 1 \equiv 0 \pmod{p}$ has exactly $p-1$ incongruent solutions modulo p (namely $x \equiv 1, 2, \ldots, p-1$) by Fermat's Little Theorem. Any such solution must be a solution to either

$$x^d - 1 \equiv 0 \pmod{p} \quad (1)$$

or

$$x^{d(e-1)} + x^{d(e-2)} + \cdots + x^d + 1 \equiv 0 \pmod{p} \quad (2)$$

(since $\mathbb{F}_p$ is a finite field, in particular an integral domain). The congruence (2) has at most $d(e-1) = p-1-d$ incongruent solutions modulo p by Theorem 5.7. Thus the congruence (1) has at least $p-1-(p-1-d) = d$ incongruent solutions modulo p. This fact combined with Theorem 5.7 implies that the congruence (1) has exactly d incongruent solutions modulo p, as desired.

<u>Exercise</u>: • Assuming that 17 has a primitive root, determine how many incongruent primitive roots there are modulo 17.

• Do the same for modulus 18.

<u>Exercise</u>:

• Prove that 3 is a primitive root modulo 43

• Use the previous part to find all incongruent integers having order 14 modulo 43.